Bipolartransistor: $B = \frac{I_C}{I_B}$ $r_{BE} = \frac{\Delta U_{BE}}{\Delta I_B} = \frac{U_T}{I_B}$ $B = \frac{\Delta I_C}{\Delta I_B}$

$D = \frac{\Delta U_{BE}}{\Delta U_{CE}}$ $I_B = I_{BS} * e^{\frac{U_{BE}}{U_T}}$ $S = \frac{I_C}{U_T} = \frac{B}{r_{BE}}$ $r_{BE} = \frac{\Delta U_{CE}}{\Delta I_C}$

$I_C = I_s * e^{\frac{U_{BE}}{U_T}} * (1 + \frac{U_{CE}}{U_A})$

Stromspiegel: $I_{C1} = I_{S1} * e^{\frac{U_{BE1}}{U_T}}$ $I_{B1} = \frac{I_{C1}}{B}$ $I_{C2} = I_{S2} * e^{\frac{U_{BE2}}{U_T}} * (1 + \frac{U_{CE2}}{U_A})$

$I_e = I_{C1} + I_{B1} + I_{B2}$ $I_a = I_{C2}$ $\mathrm{k} = \frac{\mathrm{I_a}}{\mathrm{I_e}} = \frac{\mathrm{R_1}}{\mathrm{R_2}}$

Verlustleistungen: $P_{CE} = U_{CE} * I_C$ $P_{BE} = U_{BE} * I_B$ $P_{tot} = P_{CE} + P_{BE} \approx P_{CE}$

Arbeitspunkt mit Vorwiderstand: $R_C = \frac{U_{B1} - U_{C1}}{I_C}$ $R_C = \frac{U_{B1} - U_{BE}}{I_B}$

Basisspannungsteiler: $R_1 = \frac{U_{B1} - U_{R2}}{I_1} = \frac{U_{B1} - U_{R2}}{10 * I_B}$ $R_2 = \frac{U_{BE}}{9 * I_B}$

Spannungsgegenkopplung: $R_1 = \frac{U_{CE} - U_{BE}}{I_B}$ $R_C = \frac{U_{CC} - U_{CE}}{I_B + I_B}$

Stromgegenkopplung: $R_1 = \frac{U_{B1} - U_{BE} - U_{RE}}{I_1} = \frac{U_{B1} - U_{BE} - U_{RE}}{10 * I_B}$ $R_2 = \frac{U_{BE} + U_{RE}}{9 * I_B}$ $R_E = \frac{U_{RE}}{I_B + I_C} = \frac{U_{RE}}{I_E} = \frac{U_{RE}}{I_B + \beta * I_B}$

Ersatzschaltung: $r_e = \frac{u_e}{i_e}$ $i_e = \frac{u_e}{R_1 || R_2 || R_{BE}}$ $r_e = R_1 || R_2 || r_{BE}$ $r_a = \frac{u_a}{a}$

$i_a = \frac{u_a}{R_C || r_{CE}}$ $r_a = R_C || R_{CE}$ $A_w = \frac{u_a}{u_e} = S * R_C$ $A_G = \frac{R_C}{R_E}$ $f_1 = (C_E * R_E)^{-1}$

$f_2 = (2 * \pi * C_E)^{-1} * S$

Operationsverstärker nichtinvertierend: $A = \frac{U_A}{U_E} = 1 + \frac{R_1}{R_2}$ $r_e = \infty$ $r_a = 0$

Operationsverstärker invertierend: $A = \frac{U_A}{U_E} = -\frac{R_2}{R_1}$ $r_e = R_1 r_a = 0$

Komperator: $U_- = U * \frac{R_2}{R_1 + R_2}$

Fensterkomperator: $U_{1+} = U * \frac{R_2 + R_3}{R_1 + R_2 + R_3}$ $U_{2-} = U * \frac{R_3}{R_1 + R_2 + R_3}$

Schmitt – Trigger: $U_{1+} = U * \frac{R_2 || R_3}{R_1 + R_2 || R_3} \rightarrow U_a = 0$ $U_{2+} = U * \frac{R_2}{R_1 || R_3 + R_2} \rightarrow U_a = U$